A3. $I=[0,1]$, $\Omega=\mathbb{R}$, $\alpha\in\mathbb{R}$. $A_\alpha=(\alpha-1,\alpha+1)$

(a) proof $\bigcup_{\alpha\in I}A_\alpha=(-1,2)$

$\alpha\in I \Rightarrow \alpha\in[0,1]$ $\begin{cases}\text{when } \alpha=0 & A_\alpha=(-1,1)\\ \text{when } \alpha=1 & A_\alpha=(0,2)\end{cases}$

$\forall\alpha\in[0,1]$ $A_\alpha\subset(-1,+\infty)$ and $A_\alpha\subset(-\infty,2)$

$\Rightarrow \forall\alpha\in[0,1]$ $A_\alpha\subset(-1,2) \Rightarrow \underline{\bigcup_{\alpha\in I}A_\alpha\subset(-1,2)}$

Also

$\forall x\in(-1,2)$ $\begin{cases} x\in(-1,0) \Rightarrow \alpha_1=0\in[0,1] \text{ s.t. } x\in A_{\alpha_1}=(-1,1)\\ \text{or } x\in[0,1) \Rightarrow \alpha_2=0\in[0,1] \text{ s.t. } x\in A_{\alpha_2}=(-1,1)\\ \text{or } x\in[1,2) \Rightarrow \alpha_3=1\in[0,1] \text{ s.t. } x\in A_{\alpha_3}=(0,2)\end{cases}$

$\Rightarrow x\in\bigcup_{\alpha\in\{0,1\}}A_\alpha\subset\bigcup_{\alpha\in I}A_\alpha$

$\Rightarrow \underline{(-1,2)\subset\bigcup_{\alpha\in I}A_\alpha}$

So $(-1,2)=\bigcup_{\alpha\in I}A_\alpha$.

(b) proof $\bigcap_{\alpha\in I}A_\alpha=(0,1)$

$\forall\alpha\in[0,1]$, $\begin{cases}\alpha+1\geq 1\\ \alpha-1\leq 0\end{cases} \Rightarrow (0,1)\subset(\alpha-1,\alpha+1)$ for any α $\Rightarrow \underline{(0,1)\subset\bigcap_{\alpha\in I}A_\alpha}$.

also

$\forall x\in\bigcap_{\alpha\in I}A_\alpha \Rightarrow x\in A_\alpha\ \forall\alpha\in[0,1]$

$\left.\begin{array}{l}\text{if } \alpha=0 \Rightarrow A_\alpha=(-1,1)\\ \text{if } \alpha=1 \Rightarrow A_\alpha=(0,2)\end{array}\right\} \Rightarrow \underline{\bigcap_{\alpha\in I}A_\alpha\subset(-1,1)\cap(0,2)=(0,1)}$

So $\bigcap_{\alpha\in I}A_\alpha=(0,1)$

A5. $X\equiv\{0,1\}^{\mathbb{N}}$ the set of all sequences $\{w_i\}_{i\in\mathbb{N}}$. $w_i\in\{0,1\}$.
in this set, element is certain sequence, s.t. $\{1,0,0,\dots\}$. with infinite elements taken from $\{0,1\}$.

Construct a set A s.t. the element in A is an infinite sequence with i^{th} element equal to 1 while others are 0. (look like $\{0,0,\dots,0,\underset{\uparrow\, i^{th}}{1},0,\dots\}$) for $i\in\mathbb{N}$.

So A is a true subset of X and $\exists$ 1-1 & onto function f from $\mathbb{N}$ to A s.t.
$f(i)=\{0,0,\dots,0,\underset{\uparrow\, i^{th}}{1},0,\dots\}$ So $|X|>|A|=|\mathbb{N}| \Rightarrow X$ is uncountable.

Similarly. the set $B=\{\{0\},\{1\},\{2\},\dots\}$, which is the collection of all one element subset of $\mathbb{N}$. $B\subset P(\mathbb{N})$ and $\underbrace{|B|=|\mathbb{N}|}$ So $|P(\mathbb{N})|>|B|=|\mathbb{N}|$
So $P(\mathbb{N})$ is also uncountable. ($\exists$ $f(i)=\{i\}$ 1-1 and onto function)

A.9. (a) j starts with 1

So when $n=1$ $\sum_{j=1}^{1} j^2 = 1^2 = \frac{1\times 2\times 3}{6} = 1$ holds.

Suppose when $n=k \in \mathbb{N}$, $k \geq 2$

$$\sum_{j=1}^{k} j^2 = \frac{k(k+1)(2k+1)}{6} \text{ holds.}$$

To prove when $n=k+1 \in \mathbb{N}$ $k \geq 2$ the equation also holds.

when $n=k+1$ $\sum_{j=1}^{k+1} j^2 = \sum_{j=1}^{k} j^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + \frac{6(k+1)^2}{6}$

$$= \frac{(k+1)(2k^2+k+6k+6)}{6} = \frac{(k+1)(2k^2+7k+6)}{6}$$

1 2
2 3

$$= \frac{(k+1)(k+2)(2k+3)}{6} = \frac{(k+1)(k+1+1)(2(k+1)+1)}{6} \text{ holds!}$$

So $\sum_{j=1}^{n} j^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbb{N}/\{0\}$

(b)

(i) $n=0$ $(x_1+x_2)^0 = \binom{0}{0} x_1^0 x_2^0 = 1$; $n=1$ $(x_1+x_2)^1 = x_1+x_2$ holds.

suppose $n=k$, $k \in \mathbb{N}/\{0\}$ $(x_1+x_2)^k = \sum_{r=0}^{k} \binom{k}{r} x_1^r x_2^{k-r}$ holds

when $n=k+1$ $(x_1+x_2)^{k+1} = (x_1+x_2)^k (x_1+x_2) = \left[\sum_{r=0}^{k} \binom{k}{r} x_1^r x_2^{k-r}\right](x_1+x_2)$

$$= \left[\binom{k}{0} x_1^0 x_2^k + \binom{k}{1} x_1^1 x_2^{k-1} + \cdots + \binom{k}{k} x_1^k x_2^0\right](x_1+x_2)$$

$$= \left[\binom{k}{0} x_1^1 x_2^k + \binom{k}{1} x_1^2 x_2^{k-1} + \cdots + \binom{k}{k} x_1^{k+1} x_2^0\right]$$

$$+ \left[\binom{k}{0} x_1^0 x_2^{k+1} + \binom{k}{1} x_1^1 x_2^k + \cdots + \binom{k}{k} x_1^k x_2^1\right]$$

$$= \binom{k+1}{0} x_1^0 x_2^{k+1} + \underbrace{\left(\binom{k}{1}+\binom{k}{0}\right)}_{\binom{k+1}{1}} x_1^1 x_2^k + \cdots + \underbrace{\left(\binom{k}{k}+\binom{k}{k-1}\right)}_{\binom{k+1}{k}} x_1^k x_2^1 + \binom{k+1}{k+1} x_1^{k+1} x_2^0$$

$$= \sum_{r=0}^{k+1} \binom{k+1}{r} x_1^r x_2^{k+1-r} \text{ holds}$$

So $(x_1+x_2)^n = \sum_{r=0}^{n} \binom{n}{r} x_1^r x_2^{n-r}$ for all $n \in \mathbb{N}$.

A9 (b)-(ii)

$k=1$ $(r_1=n)$ $x_1^n = \frac{n!}{n!} x_1^n = x_1^n$ equation holds

$k=2$ the case in (b)-(i), equation holds

Suppose when $k=m$ $(x_1+\cdots+x_m)^n = \sum \frac{n!}{r_1!\, r_2! \cdots r_m!} x_1^{r_1} x_2^{r_2} \cdots x_m^{r_m}$ holds

Then when $k=m+1$ $(\underbrace{x_1+\cdots+x_m}_{A} + \underbrace{x_{m+1}}_{B})^n = \sum_{i=0}^{n} \binom{n}{i} (x_1+\cdots+x_m)^i (x_{m+1})^{n-i}$

$$\Rightarrow (x_1+\cdots+x_m+x_{m+1})^n = \sum_{i=0}^{n} \binom{n}{i} \left(\sum \frac{i!}{r_1!\, r_2! \cdots r_m!} x_1^{r_1} x_2^{r_2} \cdots x_m^{r_m} \right) (x_{m+1})^{n-i}$$

$$= \sum_{i=0}^{n} \sum \frac{n!}{\cancel{i!}(n-i)!} \frac{\cancel{i!}}{r_1!\, r_2! \cdots r_m!} x_1^{r_1} x_2^{r_2} \cdots x_m^{r_m} (x_{m+1})^{n-i}$$

$$\underline{\underline{\text{set } r_{m+1}=n-i}} \sum_{r_1+\cdots+r_{m+1}=n} \frac{n!}{r_1!\, r_2! \cdots r_m!\, r_{m+1}!} x_1^{r_1} x_2^{r_2} \cdots x_m^{r_m} x_{m+1}^{r_{m+1}}$$

the equation also holds

So. $(x_1+x_2+\cdots+x_k)^n = \sum \frac{n!}{r_1!\, r_2! \cdots r_k!} x_1^{r_1} x_2^{r_2} \cdots x_k^{r_k}$

A11 $A = \{r : r \in \mathbb{Q},\ r^2 < 2\} = \{r : r \in \mathbb{Q},\ -\sqrt{2} < r < \sqrt{2}\}$

$\Rightarrow$ 2 is an upper bound of A $\quad 2 \in \mathbb{Q}$

$\Rightarrow$ A is bounded above in $\mathbb{Q}$

Suppose the sup A is denoted as S

Then $\forall\ x \in A$ $\quad x \le S \Rightarrow S \ge \sqrt{2}$

Also $\forall\ \varepsilon > 0$ $\quad \exists\ x \in A$ s.t. $S - \varepsilon \le x < \sqrt{2} \Rightarrow S < \sqrt{2} + \varepsilon$.

let $\varepsilon \to 0$ $\quad S \le \sqrt{2}$

$\Rightarrow$ $S = \sqrt{2}$ $\Rightarrow$ / has supremum $\sqrt{2} \notin \mathbb{Q}$

A12 $\{x_n\}_{n\geq 1}, \{y_n\}_{n\geq 1} \subset \mathbb{R}$.

$\forall n \quad \inf x_n \leq x_n \leq \sup x_n$. by ~ tion.

$\exists M_1 \in \mathbb{N}$, st. $n \geq M_1 \Rightarrow \underline{\lim}_{\to +\infty} (x_n+y_n) = \sup_{n\geq 1} (\inf_{j\geq n} (x_j+y_j)) \leq x_n+y_n$

$\exists M_2 \in \mathbb{N}$, st. $n \geq M_2 \Rightarrow \overline{\lim}_{\to +\infty} (x_n+y_n) = \inf_{n\geq 1} (\sup_{j\geq n} (x_j+y_j)) \geq x_n+y_n$.

$\Rightarrow \exists M \geq \max\{M_1, M_2\} \in \mathbb{N}, \Rightarrow n \geq M \quad \underline{\lim}_{\to +\infty} (x_n+y_n) \leq x_n+y_n \leq \overline{\lim}_{\to +\infty} (x_n+y_n)$

$\Rightarrow \underline{\lim}_{\to +\infty} (x_n+y_n) \leq \overline{\lim}_{\to +\infty} (x_n+y_n)$

For every $n \in \mathbb{N}/\{0\}$. $\quad \inf_{j\geq n} x_j + \inf_{j\geq n} y_j \leq x_n+y_n \leq \sup_{j\geq n} x_j + \sup_{j\geq n} y_j$

$\Rightarrow \quad \liminf_{n\to+\infty} x_n + \liminf_{n\to+\infty} y_n \leq \liminf_{n\to+\infty} (x_n+y_n)$

and $\quad \limsup_{n\to+\infty} (x_n+y_n) \leq \limsup_{n\to+\infty} (x_n) + \limsup_{n\to+\infty} (y_n)$

$\Rightarrow \liminf_{n\to+\infty} x_n + \liminf_{n\to+\infty} y_n \leq \liminf_{n\to+\infty} (x_n+y_n) \leq \limsup_{n\to+\infty} (x_n+y_n) \leq \limsup_{n\to+\infty} x_n + \limsup_{n\to+\infty} y_n$.

A17

(a) $A(x) = \sum_{n=0}^{+\infty} \frac{n}{n+1} x^n$

$R = \limsup_{n\to+\infty} \left|\frac{n}{n+1}\right|^{1/n} = 1 \Rightarrow \rho = \frac{1}{R} = 1$

(b) $A(x) = \sum_{n=0}^{+\infty} n^p x^n \quad p \in \mathbb{R}$ fixed.

$R = \limsup_{n\to+\infty} |n^p|^{1/n} = \limsup_{n\to+\infty} n^{\frac{p}{n}} = 1 \Rightarrow \rho = 1$

(c) $A(x) = \sum_{n=0}^{+\infty} \frac{1}{n!} x^n$

$R = \limsup_{n\to+\infty} \left|\frac{1}{n!}\right|^{\frac{1}{n}} = 0 \Rightarrow \rho = +\infty$. i.e. this power function converges everywhere.

A18

(a) $f(x)=\frac{1}{1-x}$, $a=0$, $I\equiv(-1,1)$

$f'(x)=(-1)^2\cdot(1-x)^{-2}$; $f''(x)=(-2)(1-x)^{-3}(-1)=(-1)^2 2(1-x)^{-3}$; $f^{(3)}(x)=2\times(-3)\times(1-x)^{-4}(-1)$ ---

$\Rightarrow f^{(n)}(x)=n!\,(1-x)^{-(n+1)}$

$\Rightarrow P_n^a(x)=\sum_{k=0}^{n}\frac{f^{(k)}(a)}{k!}(x-a)^k$ at $a=0$

$\Rightarrow P_n(x)=\sum_{k=0}^{n}\frac{k!}{k!}(x)^k \Rightarrow A_n=\sum_{n=0}^{+\infty}x^n$ is a power series with $a_n=1$

$\Rightarrow \limsup_{n\to+\infty}|1|^{\frac{1}{n}}=1 \Rightarrow \rho=1 \Rightarrow$ series converges on $(0-1,0+1)$

$\Rightarrow$ the power series converges in $I=(-1,1)$

(b) $f(x)=1+x+x^2$, $a=2$, $I=(1,3)$

$f'(x)=2x+1$, $f''(x)=2$ $f^{(3)}(x)=0$. $f^{(k)}(x)=0$ $k\geq3$

$\Rightarrow P_n^a(x)=\sum_{k=0}^{n}\frac{f^{(k)}(a)}{k!}(x-a)^k$ $a=2$

$=7+(4+1)(x-2)+\frac{1}{2}\times2\times(x-2)^2$

$=7+5x-10+x^2-4x+4$

$=x^2+x+1$

(c) $f(x)=\begin{cases}e^{-\frac{1}{x^2}} & \text{if } |x|<1 \quad x\neq0\\ 0 & \text{if } x=0\end{cases}$

(i) $f'(x)=\lim_{x\to0}\frac{f(x)-f(0)}{x-0}=\lim_{x\to0}\frac{e^{-\frac{1}{x^2}}}{x}=0$

So $f'(x)$ is differentiable at 0, and it equals to 0.

suppose $f(x)$ is k^{th} order differentiable at 0, and equals to 0

then $f^{(k+1)}(x)=\lim_{x\to0}\frac{f^{(k)}(x)-f^{(k)}(0)}{x-0}$

where $f^{(k)}(x)\Big|_{x\neq0,\ |x|<1}=\sum_{i=0}^{k-1}\frac{e^{-\frac{1}{x^2}}\times(\text{some constant})}{x^{3k-2i}}$

⋮

$$\Rightarrow \lim_{x\to 0} \frac{f^{(k)}(x) - f^{(k)}(0)}{x-0} = \lim_{x\to 0} \frac{f^{(k)}(x)}{x} = \lim_{x\to 0} \sum_{i=0}^{k+1} \frac{e^{-\frac{1}{x^2}} * (\text{some constant.})}{x^{3k-2i-1}}$$

$$= 0$$

$\Rightarrow$ $f(x)$ is $(k+1)^{th}$ order differentiable at 0 and equals to 0

$\Rightarrow$ $f^{n}(0)$ is always 0.

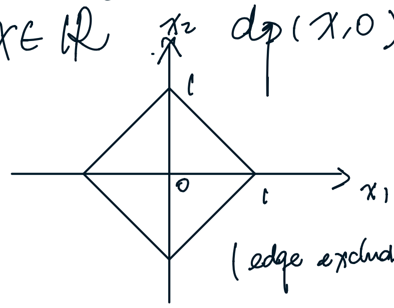
(ii) $$P_n^{a=0}(x) = \sum_{k=0}^{n} \frac{0}{k!}(x-0)^k = 0$$

So the series converges, but it converges to 0, not to $f(x)$. on $(-1,1)$

(i.e. it converges to f only at $x=0$)

A24 (c) $x \in \mathbb{R}^2$ $d_p(x,0) = \sqrt[p]{(x_1-0)^p + (x_2-0)^p}$

$p=1$



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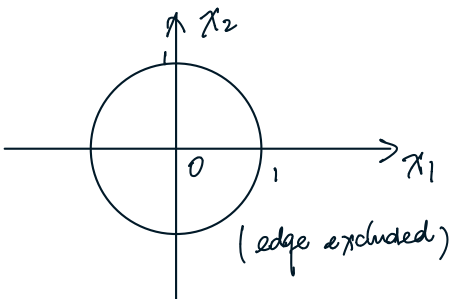
$|x_1 + x_2| < 1$

$p=2$



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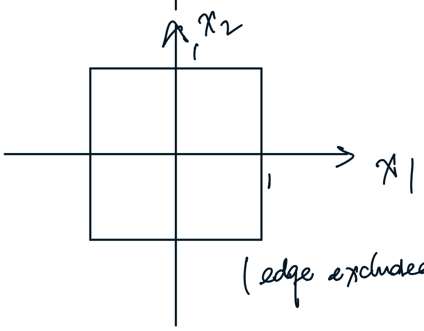
$\sqrt[2]{x_1^2 + x_2^2} < 1$

$p=\infty$



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